

REVIEW OF:
**SLOPE-AWARE LPV-MPC PATH TRACKING OF A DUAL-STEERING-WHEEL AGV WITH TIME-LAG-
ENHANCED MODERNTCN PERCEPTION**

Dear Authors,

Your efforts are commendable and praiseworthy. Please note some points.

Main Contribution:

The manuscript develops a simulation-based perception-to-control framework for path tracking of a diagonal dual-steering-wheel AGV under longitudinal road-slope variations. Its central contribution is the integration of (i) a lightweight ModernTCN-inspired history-based slope estimator using a 22-dimensional proprioceptive feature bank and multi-scale temporal-difference channels, (ii) a bounded uncertainty-adaptive fusion mechanism in which a qualified IMU observer provides an auxiliary correction to the learned estimate, and (iii) a slope-scheduled LPV-MPC in which one conditioned slope signal is used consistently to update the prediction model, measured-disturbance term, feedforward compensation, weights, and constraints. The paper also provides a relatively extensive simulation protocol comparing zero-slope, truth-driven, observer-based, learned-estimator, and fused scheduling configurations.

Major Comments:

- 1) In the Abstract, the manuscript states that the proposed method addresses degradation of control accuracy and stability and later implies improved stability. However, no formal closed-loop stability, recursive-feasibility, robustness, or bounded-error analysis is provided. Since Section VI explicitly leaves formal stability and error-bound analysis to future work, all stability-related claims in the Abstract and elsewhere should be removed or substantially qualified unless a rigorous analysis is added.
- 2) In Section I (Introduction), the novelty needs to be positioned more sharply against existing learning-assisted MPC and road-slope-estimation literature. The current three contributions combine several modifications at once (feature engineering, temporal differencing, a simplified ModernTCN backbone, observer fusion, and slope-scheduled LPV-MPC), making it difficult to determine which elements are fundamentally new and which are task-specific engineering choices. A clearer novelty table or structured comparison with the closest prior works would substantially improve the manuscript.
- 3) In the Introduction, the discussion of observer-assisted estimation and robust control for mobile robotic systems could be broadened. In addition to road-slope-specific observers, the authors may consider briefly discussing related estimation–control frameworks in which Extended Kalman Filters are

integrated with robust control schemes for mobile robots. Such studies can help better position the proposed perception–control architecture with respect to established observer-based robotic control approaches. One relevant example is: “Sliding Mode Controller via Extended Kalman Filters for Mobile Robot”.

4) In the Introduction, the literature review on path-tracking control strategies could be broadened. Since model predictive control can be regarded as a receding-horizon optimal control strategy, it would be useful to briefly discuss its relationship with other optimal and robust control approaches. In particular, constrained optimal/robust control and sliding-mode control have been extensively investigated for robotic systems under constraints, uncertainties, disturbances, and faults. Such a discussion would better position the proposed LPV-MPC framework within the broader field of optimal and robust control and clarify its advantages in prediction, constraint handling, and adaptation to slope-dependent operating conditions. The authors are encouraged to consider the following relevant studies:

- Constrained Control of Fully-Constrained Cable-Driven Parallel Robots With Elastic Cables: An Optimal Robust Approach
- Robust Optimal Constrained Control of Fully-Constrained Cable-Driven Parallel Robots Based on GSDRE
- Observer-Based Active Fault-Tolerant Sliding Mode Control for Fully-Constrained CDPRs

5) In Section II-A (AGV Model), the treatment of the passive RF and LR wheels requires stronger physical justification. Their normal loads and rolling resistance are retained, while their lateral-force and yaw-moment contributions are neglected. For path tracking with significant steering and yaw motion, omitting the lateral dynamics of two supporting wheels may materially alter the lateral force balance, yaw dynamics, and sideslip behavior. The authors should justify the passive-wheel mechanical configuration and demonstrate that this simplification does not bias the conclusions.

6) In Section II-A, Eq. (7) appears to omit the steady front-rear normal-load redistribution caused by the tangential gravity component acting at the center-of-gravity height on an inclined road. The text attributes this omission to the exclusion of body-pitch dynamics, but a rigid vehicle on a constant slope can still exhibit a static axle-load shift proportional to $m g h_{cg} \sin(\theta)/L_w$. This issue directly affects the normal loads used in Eqs. (8)-(9), tire-force limits, and therefore the claimed slope sensitivity of the plant. The load model should be re-derived or the approximation should be explicitly justified and quantified.

- 7) In Section II-A, the force-coordinate treatment in Eqs. (9)-(11) needs careful verification. The longitudinal drive force is rotated into the body frame through $R(\delta_i)$ in Eq. (10), whereas the lateral tire forces in Eq. (9) appear to enter the yaw moment and sideslip dynamics without an equivalent steering-angle rotation. This is particularly concerning because Table 1 allows steering angles up to 90 degrees. The authors should provide a consistent wheel-to-body force transformation and derive the complete yaw moment from $r_i \times F_i$ for both longitudinal and lateral tire-force components.
- 8) In Section II-A, Eq. (4) uses a voltage-limited current expression based on $V_{bus} - k_e \omega_m$ and then applies a symmetric current saturation of $\pm I_{eff}$. This appears to use a motoring-side voltage limit for both positive and negative current. Regenerative/braking operation generally has a different voltage constraint. Since the controller allows negative F_{cmd} , the electrical model should be made sign-consistent or the simplifying assumption should be stated and its effect assessed.
- 9) In Section II-A and Table 1, several quantities required by the implemented plant are not reported among the listed physical parameters, including the motor electrical parameters used to compute τ_e , V_{bus} , k_e , R_m , and L_m , as well as thresholds such as ω_{min} and v_{low} and the internal filter constants collected in ζ_f . Because the same plant is used for data generation, LPV linearization, and closed-loop evaluation, these missing values prevent independent reproduction of the simulations.
- 10) In Section II-B and throughout the methodology, the same nonlinear plant revision and parameter set are used for synthetic data generation, LPV database construction, and closed-loop evaluation. This creates a highly matched simulation environment and risks an inverse-crime-type evaluation in which the estimator and controller are validated on the same modeling assumptions that generated the data. The paper should include systematic model-mismatch tests (mass, inertia, tire stiffness, friction, motor constants, sensor biases, load changes, etc.) or clearly limit the claims to nominal matched-model simulation.
- 11) In Section III-A, the proposed network is substantially simplified relative to the original ModernTCN, as acknowledged in Table 2. Because patch embedding, the multi-stage hierarchy, re-parameterized kernels, and grouped ConvFFN structure are removed, the manuscript should justify the use of the name 'ModernTCN' rather than 'ModernTCN-inspired' or 'ModernTCN-derived'. More importantly, an ablation should separate the benefit of the backbone choice from the benefit of the engineered feature and delta representation.

REVIEW OF:
**SLOPE-AWARE LPV-MPC PATH TRACKING OF A DUAL-STEERING-WHEEL AGV WITH TIME-LAG-
ENHANCED MODERNTCN PERCEPTION**

12) In Section III-A, Eq. (19) is not reproducible as written. The precise definitions of L_{θ} , L_{main} , L_{turn} , and R_{θ} are not provided, and the numerical values of λ_{θ} and λ_{turn} are absent. The flat-road and bounded-error penalties contained in R_{θ} also require explicit formulas and coefficients. These details are necessary to reproduce the reported estimator performance.

13) In Section III-A and Section III-D, c_{main} is taken from the maximum softmax probability of an auxiliary operating-condition classifier and then used to calibrate ModernTCN uncertainty for fusion. A maximum softmax score is not automatically a calibrated uncertainty measure. The authors should report the calibration method, calibration error/reliability diagram, and the exact mapping from c_{main} to $P_{M,k}$; otherwise the uncertainty-adaptive fusion is not sufficiently defined.

14) In Section III-B, several derived features depend on numerical differentiation of wheel-speed signals, for example $a_{x,w,k}$ and the filtered acceleration terms in Eq. (21). Such operations can strongly amplify measurement noise. Because the study is entirely simulation-based, the manuscript should specify the wheel-speed quantization/noise, current-sensor noise, steering-sensor noise, gyro noise, bias, vibration, and filtering models used during both training and closed-loop evaluation. Without this information, the difficulty of the slope-estimation task cannot be assessed.

15) In Section III-C, the manuscript states that lag offsets from $\{1, 2, 4, 8, 16, 32, 64\}$ and several representation variants were screened, but only the final ModernTCN-22D versus ModernTCN-Delta comparison is reported in the results. The complete lag/representation ablation used to select $\{1, 2, 4\}$ should be reported, preferably on validation data only, to demonstrate that the selected delta bank is not the result of undocumented post-selection.

16) In Section III-C, the authors state that $N_h = 128$ (1.28 s) was fixed from a characteristic response-time calculation and that the corresponding calculation is provided in the experimental setup. I could not find that calculation in Section IV. The manuscript should explicitly show the derivation, the representative speed/slope/effective-inertia values used, and a sensitivity study for the window length.

17) In Section III-D, the qualified observer is described only through generic maps $f_{\text{obs}}(\cdot)$, $h_{\text{obs}}(\cdot)$, $G_{\theta}(\cdot)$, and $F(\cdot)$ in Eqs. (28)-(30). This is insufficient for reproduction. The complete observer state vector, propagation equation, gravity

observation equation, Jacobians (if applicable), covariance update, quality-score definitions, and road-aligned slope conversion should be provided.

18) In Section III-D, the fusion variables $P_{M,k}$, $P_{O,k}$, and C_k are central to Eq. (31), but the manuscript does not provide the exact regime definitions, the stored error-moment values, or the formulas by which the ModernTCN confidence and observer quality score modify those moments online. Since the claimed contribution is an 'uncertainty-adaptive' fusion interface, these equations and calibration procedures should be fully specified.

19) In Section III-D, the fusion logic contains many empirically selected thresholds (helpfulness 0.6, slope entry/exit 0.5/0.3 deg, minimum innovation 0.25 deg, innovation gate 2 deg, NIS thresholds 9/25, correction bound 0.5 deg, rate bound 5 deg/s, and a 1.3 deg scheduling deadband). No sensitivity or robustness analysis is provided for these choices. In particular, Eq. (32) produces a discontinuous transition from a nonzero downweighted correction immediately below $NIS = 25$ to zero immediately above it. The authors should justify these thresholds and demonstrate that the conclusions are not sensitive to manual tuning.

20) In Section III-D, the phrase 'exact fallback to the primary estimate' should be defined more carefully. Eq. (33) indeed sets the fused estimate equal to the ModernTCN estimate when a gate fails, but the signal subsequently passes through a 1.3 deg deadband, amplitude clipping, and the scheduling filter in Eq. (34). Therefore, the final θ_{sch} supplied to the controller is not necessarily identical to the raw or deployment-filtered ModernTCN estimate. The manuscript should distinguish exact fusion-stage fallback from the final conditioned scheduling signal.

21) In Section III-E, the role of slope as both a scheduling variable and a measured disturbance needs a clearer derivation to rule out double counting. Equation (38) schedules A_d , B_d , and E_d with θ_{sch} ; Eq. (43) also includes the slope-deviation channel through E_d , while Eqs. (41) and (45) add slope-dependent feedforward. The authors should explicitly define the operating-point value $\bar{\theta}$ used online and show algebraically how the disturbance deviation, scheduling interpolation, and feedforward terms combine without duplicating the same slope effect.

22) In Section III-E, there is a notation/interpretation ambiguity between the path-error state and the MPC reference. Equation (35) defines x_e entirely as tracking errors and Eq. (38) uses $C_d = I_4$, yet Eq. (46) penalizes $y_{\{k+i|k\}} - y_{ref_{\{k+i\}}}$. If y is already the error vector, the natural reference is zero. The

authors should define y_{ref} explicitly and ensure that the cost is consistent with the error-state formulation.

23) In Section III-E, trilinear interpolation among 4125 local models, frozen scheduling over the prediction horizon, slope-dependent weight/constraint scheduling, and soft output constraints are introduced without any analysis of recursive feasibility or closed-loop stability. At minimum, the authors should discuss the implications of interpolation and scheduling-rate changes on feasibility and boundedness, report constraint-violation statistics, and avoid stability language unless a formal result is provided.

24) In Table 4 and Section III-E, the input limits are internally confusing. F_{cmd} is listed with an input constraint of $[-600, 600]$ N but also an 'online envelope' of $[-720, 720]$ N; similarly, ω_{cmd} is listed as $[-1.2, 1.2]$ rad/s with an online envelope of $[-1.44, 1.44]$ rad/s. It is unclear which bounds are hard physical limits and which are scheduled limits. The hierarchy of plant saturation, MPC hard bounds, and scheduled envelopes must be stated unambiguously.

25) In Section IV-C and Algorithm 2, the split is run-disjoint but apparently not template-disjoint. The 102 runs are generated from 51 route templates with two realizations per template, and the algorithm splits run identifiers 71/15/16. If two realizations of the same template can appear in different splits, the test set is not structurally independent from training. The authors should use a template-level split, or demonstrate that template identities are also disjoint across training, validation, and test sets.

26) In Section IV, the paper is titled and written as a control/perception study for an AGV, yet all training and evaluation evidence is synthetic. The section heading 'Experimental Setup' may therefore be misleading. More importantly, no hardware-in-the-loop or physical-vehicle validation is provided. At a minimum, the paper should include domain-gap stress tests with realistic parameter uncertainty, sensor bias/noise, disturbances, and unseen route/load conditions; otherwise the scope of the claims should be restricted to simulation.

27) In Section IV-F, the statistical interpretation needs revision. The ten neural-network seeds are repeated training realizations on the same dataset and are not independent samples from the deployment population. Likewise, the closed-loop benchmark contains only six route geometries. Bootstrap confidence intervals over seeds (or seed-first hierarchical resampling) primarily quantify training stochasticity under these particular simulated routes, not generalization to new routes or physical vehicles. The statistical claims should be reframed and,

if possible, expanded using independently generated route templates as the statistical units.

28) In Section IV-F and Section V, the composite input-increment index J_Delta_u is never given an explicit mathematical definition or normalization. This is especially important because ΔF_cmd and $\Delta \omega_cmd$ have different physical units and very different numerical scales. The very large zero-slope value (approximately 696) versus sub-unit values for slope-informed cases cannot be interpreted without knowing the exact scaling, aggregation, and horizon normalization. Please provide the formula and units/dimensionless normalization.

29) In Section V-A, the performance gap between zero-slope scheduling and slope-informed scheduling is extraordinarily large: lateral RMS changes from approximately 1.48 m to approximately 0.02 m, and J_Delta_u from approximately 696 to approximately 0.5. This raises a basic sanity-check question about whether the zero-slope controller is entering saturation, becoming infeasible, or experiencing a small number of near-failure cases. Route-resolved time histories, constraint-activity plots, solver status, and failure diagnostics should be shown to establish that the comparison reflects slope information rather than a baseline implementation pathology.

30) In Section V-A, the qualified observer alone performs very close to the truth-driven scheduling reference in the aggregate metrics. Given that the observer is driven by a simulated six-axis IMU and is one of the signals later used for fusion, the authors should report the observer's standalone slope-error statistics and the exact synthetic IMU disturbance model. Otherwise, it is difficult to judge whether the observer baseline is unrealistically favorable because it is generated from the same nominal plant.

31) In Section V-E and Table 11, the confidence interval for J_Delta_u includes zero [-0.01349, 0.93333]. Therefore, the 27.61% reduction in the mean should not be presented as statistically supported in the same sense as the lateral- and heading-error improvements. The text is partly cautious, but the Abstract, Discussion, and Conclusion should consistently distinguish 'lower mean' from an interval-supported improvement.

32) In Section V-E and Figure 8, one J_Delta_u effect is reported as 12.72 and plotted off-scale. This outlier is large relative to the other effects and to the overall mean. The authors should identify the corresponding route/seed, explain the underlying controller behavior, and report robust summaries (median/IQR or

trimmed sensitivity) to show whether the mean reduction is driven materially by this single case.

33) In Section V-F, the real-time claim is incomplete. Table 12 measures only the core estimator inference time, while the online loop also includes feature construction, normalization/delta updates, observer propagation/update, fusion logic, slope conditioning, LPV interpolation, weight/bound scheduling, and most importantly the MPC quadratic-program solve. With $T_s = 10$ ms, end-to-end latency and worst-case or at least p99 execution time of the complete closed loop are required before real-time feasibility can be claimed.

Minor Comments:

34) In Section I (Introduction), the description of the diagonal dual-steering-wheel arrangement should be made more explicit before the contribution list, because the unusual LF/RR active-wheel configuration is important for interpreting the subsequent kinematics and force model.

35) Throughout the manuscript, several main section headings are followed immediately by a subsection heading without any introductory text. A main section should preferably begin with a brief introductory paragraph that explains the purpose, scope, and organization of the section and provides a logical transition from the preceding section before introducing the first subsection. The authors are therefore requested to revise the manuscript structure accordingly.